EECSE6694 Project Proposal: Benchmarking LLM Circuit Understanding

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Motivation:

As three electrical engineering students, we have tried using LLMs as tutors to help us understand and solve circuit questions. However, we have found their responses difficult to rely on when the questions include schematic diagrams. A model may misread how components are connected and then give a convincing explanation based on an incorrectly interpreted circuit. This makes it hard to use the explanation for learning, especially when we are unsure of the correct solution ourselves.

These experiences motivated a broader question: how reliably can LLMs understand circuits, and how does the input format affect their performance? We propose a reproducible benchmark that presents the same circuits as schematic images and verified SPICE netlists. We will evaluate whether models can identify circuit structure, explain circuit functions, and predict electrical behavior. Comparing these tasks across input formats will help us investigate where errors arise. Our goal is to provide a reusable evaluation framework that can inform both educational uses and engineering applications.

What We Plan to Do

- **Run a pilot evaluation.** Test two or three accessible multimodal LLMs on approximately 10 to 15 circuits. The same models will receive both image and netlist inputs. We will use the pilot to refine the questions and scoring rules, then set the final dataset size based on annotation time and model access costs.
- **Define circuit coverage and difficulty.** Begin with resistor networks and first-order RC circuits. Use the provisional reasoning levels below, and record component, branch, and loop counts separately. We will finalize the coverage during the pilot.
- **Prepare matched input formats.** Provide a schematic image and a verified SPICE netlist for each circuit. Both will contain the same component values, source information, and input and output definitions. The netlist input will be checked independently, rather than extracted by the model being tested. We will remove answer-revealing titles, comments, and simulation results. A plain-language circuit description may be added later as a third format.
- **Prepare verified reference answers.** For each circuit, record its components, connectivity, function, and numerical answers. Cross-check applicable results with hand calculations and LTspice or ngspice under the same component assumptions and operating conditions. Specify units, polarity conventions, and initial conditions where needed. The simulator will generate reference answers; the baseline models will not have access to it.
- **Test several aspects of understanding.** Include questions about components and connections, circuit function, numerical outputs, and the effect of changing a component value. Ask the same questions across input formats where applicable. For example, a voltage-divider question can ask for the output voltage and how it changes when one resistor increases. Connectivity questions measure how models read the input; behavior questions test how they use that information.

Run controlled model comparisons. Use fixed prompts, answer formats, and comparable response budgets, with a fresh session for each test question and input condition. Record model versions, dates, settings, and raw responses. Finalize prompts and scoring on development circuits before evaluating a separate test set. Keep all image and parameter variants of a base circuit in the same split, and report any circuit families shared across the splits.

Score performance and inspect errors. Report results separately for structure, function, and numerical reasoning. Check connectivity after normalizing arbitrary node labels, use a predefined rubric for functional answers, and set absolute and relative error tolerances for numerical answers before final testing. Review failures for component, connectivity, netlist-reading, reasoning, and calculation errors; retain an unclear category when the evidence is insufficient. Better netlist performance may indicate a visual interpretation problem, but it will not by itself prove the cause of an error.

Investigate selected visual factors. Choose one or two factors suggested by the pilot, such as junction-dot size or layout density. Change one factor at a time while preserving the electrical circuit and checking that the intended connections remain unambiguous. This will provide a focused analysis of visual sensitivity without requiring every possible factor combination.

Expected Outcomes

The main outcome will be a reusable benchmark package containing schematic images, verified netlists, questions, reference answers, circuit metadata, and evaluation scripts. It will also include the prompts, model settings, simulation setup, and instructions needed to repeat the evaluation. We will report the number of base circuits separately from image variants and question counts.

Our results will compare the selected models across input formats, question types, and reasoning levels, and document representative errors and selected visual sensitivity experiments. We will report the scope and limitations of the dataset. The framework should help others evaluate additional models, although hosted model updates and response variation may prevent identical scores on every rerun.

Optional Extensions

If the core benchmark is complete and time permits, we will test one focused improvement, such as enlarged image regions or a component-and-connection checklist. Any method will be developed on development examples and evaluated on the held-out test circuits. Circuit design can be a separate extension, such as proposing a voltage divider that meets a specified output target and checking it through simulation. The core deliverable will remain the circuit-understanding benchmark.

Variable	Planned comparison	Priority
Input format	Schematic image versus verified SPICE netlist of the same circuit.	Core
Model	Two or three multimodal LLMs tested under both input conditions.	Core
Circuit and question difficulty	Basic DC relationships, multi-step DC analysis, and first-order RC behavior.	Core
Task	Structure recognition, function identification, numerical prediction, and parameter changes.	Core
Visual features	Select one or two: wire crossings, junction dots, resolution, density, drawing style, or component-symbol style.	Secondary
Additional methods or inputs	A structured checklist, enlarged image regions, or a plain-language circuit description.	Optional

Baseline evaluation: separate image-only and verified-netlist-only conditions, each with the fixed prompt and no simulator or external tool access. These are the main reference conditions for any optional improvement experiment.

References

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